

Behavioral Analytics for Customer Retention: The Role of Engagement and Product Utilization in Banking

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Abstract— This study uses a behavioral analytics approach to examine customer retention in the banking industry, giving priority to product utilization and customer engagement over conventional financial metrics. By using exploratory data analysis to analyze a dataset of 10,000 customers, the study looks at engagement trends, churn distribution, and the impact of product adoption. To capture intricate retention dynamics, it introduces new retention KPIs: Relationship Strength Index, Product Depth Index, High-Balance Disengagement Rate, and Engagement Retention Ratio. An interactive Streamlit dashboard was created to show insights and pinpoint customer segments that pose a high risk. The findings show that while financial indicators by themselves are insufficient to predict loyalty, engagement and product use are important factors that influence retention. These results provide banks with practical ways to improve long-term relationships, maximize product offerings, and boost customer engagement.

Keywords— *Customer Retention, Customer Churn, Banking Analytics, Product Utilization, Engagement Analytics, Data Visualization*

I. INTRODUCTION

In today's competitive banking world, keeping customers is very important for banks. Banks need to know what makes customers stay with them for a long time because keeping current customers is often much cheaper than getting new ones. In the past, many banks used demographic and financial data, like income level, account balance, or customer profile, to figure out how likely it was that a customer would leave.

These metrics might not, however, adequately convey the quality of a client's relationship with the bank. According to recent research, behavioral elements like product usage and customer engagement offer more profound insights into retention trends. Consumers who actively utilize banking services, engage with digital platforms on a regular basis, and

embrace a variety of financial products typically build stronger bonds with their bank.

Many banks find it difficult to convert the vast amounts of customer data they have access to into useful insights about consumer behavior. Because of this, retention tactics frequently stay general and might not successfully target the clients who are most likely to leave.

This project focuses on using behavioral analytics to analyze customer retention by looking at indicators of financial commitment, product usage, and engagement patterns. The goal is to pinpoint the main behavioral factors that influence customer loyalty and offer data-driven insights that can help with more focused and successful retention tactics.

II. PROBLEM STATEMENT

Despite having access to vast amounts of customer data, banks still find it difficult to turn this information into useful insights regarding customer attrition. Conventional retention strategies frequently place a strong emphasis on demographic profiles or financial metrics, which might not adequately capture the behavioral cues that point to possible churn or declining engagement.

The Banks frequently lack the analytical frameworks necessary to comprehend how product usage and customer engagement affect retention results. Because of this, it is challenging to measure how behaviors like digital activity, transaction frequency, or the use of several banking products affect long-term client loyalty.

Financial institutions also frequently have trouble spotting high-value clients who might be in danger of leaving even though they seem to be financially secure. These clients may become "silent churn risks" in the absence of appropriate behavioral analysis, making it more difficult for banks to step in with focused retention tactics.

By investigating the connection between customer engagement, product utilization, and churn behavior, this project tackles these issues. The objective is to produce data-driven insights that assist banks in better identifying at-risk clients and creating engagement-focused retention strategies that are more successful.

III. RESEARCH OBJECTIVES

A. Primary Objectives

- 1)Assess the impact of customer engagement on churn risk by analyzing factors such as transaction frequency, digital activity, and interaction levels.
- 2)Evaluate how product utilization influences customer retention, particularly the effect of multi-product adoption on long-term loyalty.
- 3)Identify high-value customers who show signs of disengagement and may be at risk of churn despite strong financial profiles.

B. Secondary Objectives

- 1)Support the development of data-driven, engagement-focused retention strategies instead of generic approaches.
- 2)Provide insights to improve product bundling and cross-selling decisions based on customer usage patterns.
- 3)Detect early warning signals of churn among premium customers to enable timely and targeted intervention.

IV. DATASET DESCRIPTION

The study's dataset, which includes data on 10,000 banking clients, combines financial indicators, behavioral engagement metrics, and demographic characteristics. A data-driven analysis of retention trends is made possible by this dataset, which offers a thorough view of customer profiles and their interactions with banking services.

The key variables included in the dataset are as follows:

Variable	Description
CustomerId	Unique identifier assigned to each customer
CreditScore	Indicator of the customer’s creditworthiness
Geography	Customer location (France, Spain, Germany)
Gender	Customer gender (Male/Female)
Age	Age of the customer
Tenure	Number of years the customer has been associated with the bank
Balance	Account balance maintained by the customer
NumOfProducts	Number of banking products used by the customer
HasCrCard	Indicates whether the customer owns a credit card
IsActiveMember	Reflects whether the customer is actively engaged with the bank
EstimatedSalary	Estimated annual income of the customer

Variable	Description
Exited	Churn indicator showing whether the customer has left the bank

Table 1. Dataset Description

The "Exited" variable, which is the target variable for determining customer churn, is the main focus of this analysis. The study seeks to identify the critical elements affecting customer retention and churn behavior by examining trends in engagement, product usage, and financial attributes.

V. ANALYTICAL METHODOLOGY

This project uses behavioral insights to assess customer retention using an organized and methodical analytical approach. The process is intended to convert unprocessed client data into useful insights that can facilitate data-driven decision-making in a banking setting.

A. Data Ingestion and Validation

The Pandas library was used to import the dataset into a Python environment for data analysis and manipulation. Verifying data types, looking for missing or inconsistent values, and guaranteeing the integrity of important variables were among the first steps in data validation. To ensure accuracy and consistency across records, binary indicators like IsActiveMember, HasCrCard, and Exited were specifically validated.

B. Engagement Classification

Based on their activity levels and product usage patterns, customers were divided into different engagement profiles. Customers with limited product adoption despite being active, inactive or disengaged customers, and high-balance customers with low engagement were among these segments. A more detailed understanding of behavioral patterns and their connection to churn was made possible by this classification.

C. Product Utilization Analysis

The analysis looked at how a customer's usage of various banking products affects retention. Churn rates were computed for each group of customers based on the degree of product adoption. This made it easier to determine whether increased product usage is associated with lower churn risk and higher customer loyalty.

D. Financial Commitment vs Engagement Analysis

Key financial indicators like account balance and estimated salary were examined in conjunction with engagement status to evaluate the relationship between financial strength and behavioral activity. This method highlighted potential "silent churn" segments by identifying high-value clients who might still be at risk of churn due to low engagement.

E. Retention Strength Assessment

Several dimensions, such as credit card ownership, product utilization, and customer engagement, were combined to create a composite Relationship Strength Score. This score helped

prioritize customers for retention efforts and gave a comprehensive picture of the strength of the customer-bank relationship.

F. Dashboard Development

Streamlit was used to create an interactive dashboard that shows important retention insights. Business users and decision-makers can more easily access the analysis thanks to the dashboard's dynamic exploration of customer segments, churn patterns, and engagement metrics through filters and interactive elements.

VI. EXPLORATORY DATA ANALYSIS

A. Churn Distribution

According to the dataset's preliminary analysis, about 20% of customers have left, while the remaining 80% have been kept. Despite making up a small portion of the customer base, churn is important from a business standpoint because even slight increases in churn can result in significant revenue loss. The significance of precisely identifying at-risk clients for focused retention initiatives is further highlighted by this class disparity.

B. Engagement vs churn

One of the most important variables in determining retention results was found to be customer engagement. According to the analysis, compared to active members, inactive customers are much more likely to churn. In fact, inactive customers exhibit almost twice the churn probability, highlighting the crucial role that regular engagement and activity play in fostering customer loyalty. This result emphasizes how important it is for banks to keep a careful eye on engagement metrics as early warning signs of churn risk.

C. Product Utilization Impact

The number of products used and customer retention have a nonlinear relationship, according to the product utilization analysis. The lowest churn rates were shown by customers who used two banking products, indicating that moderate product adoption improves the customer-bank relationship. Customers who had three or more products, however, had comparatively higher churn rates, which could point to problems like mismatched products, low perceived value, or poor cross-selling tactics. This emphasizes how crucial it is to match product offerings with consumer demands rather than concentrating only on growing the number of products.

D. High-Value Disengaged Customers

A unique group of consumers with large account balances and high projected salaries but low levels of engagement was found. These clients show signs of "silent churn risk," where disengagement could eventually result in attrition, despite having solid financial profiles. Because losing high-value clients can have a disproportionate effect on overall profitability, this segment is especially important for banks.

E. Relationship Strength Analysis

A composite relationship strength score was created based on credit card ownership, product usage, and engagement status in

order to gain a better understanding of customer loyalty. Customers with moderate relationship strength scores, which reflect a balanced level of engagement and product usage, show the highest retention stability, according to the analysis. Customers with low relationship scores, on the other hand, have the highest churn probability, highlighting the significance of increasing both engagement and product adoption to enhance retention results.

VII. KEY PERFORMANCE INDICATORS

A set of key performance indicators (KPIs) was created in order to assess the factors that influence customer retention. These metrics allow for a more organized and data-driven evaluation of retention dynamics by quantifying the effects of engagement, product usage, and financial behavior on churn.

1)Engagement Retention Ratio : The difference in churn rates between active and inactive customers is measured by this metric. It serves as a powerful gauge of the success of engagement-driven tactics and emphasizes the degree to which customer engagement affects retention.

2)Product Depth Index: The Product Depth Index assesses the correlation between a customer's likelihood of retention and the quantity of banking products they use. It assists in determining whether higher product adoption boosts customer loyalty or whether diminishing returns are seen above a particular threshold.

3)High-Balance Disengagement Rate: Customers with large account balances who show low levels of engagement are identified by this KPI. It is especially helpful for identifying high-value clients who might be at risk of leaving even though they seem financially secure, allowing for focused intervention tactics.

4)Credit Card Stickiness Score: The Credit Card Stickiness Score evaluates whether having a credit card improves customer retention. It offers information on how particular products can improve enduring client relationships and lower the likelihood of churn.

5)Relationship Strength Index: The Relationship Strength Index is a composite metric that provides an overall assessment of customer loyalty by combining product utilization and customer engagement. Customers can be divided into groups according to the strength of their relationships thanks to this KPI, which also helps focus retention efforts on at-risk groups.

VIII.DASHBOARD IMPLEMENTATION

Using Streamlit, an interactive dashboard was created to operationalize the analytical results and make them available for business decision-making. The dashboard offers a centralized interface for investigating customer retention trends, allowing for dynamic analysis of customer risk segments, product utilization, and engagement.

The following link will take you to the interactive dashboard created for this project:

Dashboard Link: [\[Bank Retention Intelligence · Streamlit\]](#)

1)Engagement vs Churn Analysis:

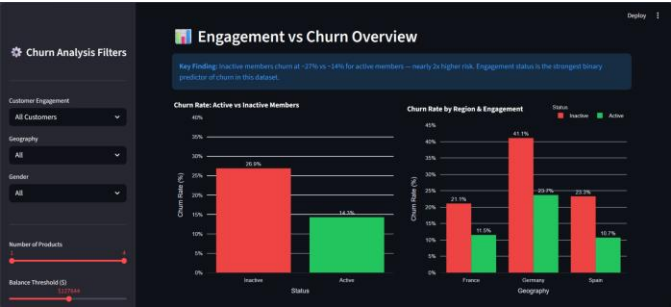


Fig. 1. Engagement vs churn overview

This graphic illustrates how customer engagement and churn are related. It demonstrates unequivocally that inactive customers have much higher churn rates than active members, highlighting engagement as a crucial factor in customer retention.

2)Product Utilization Impact Analysis:

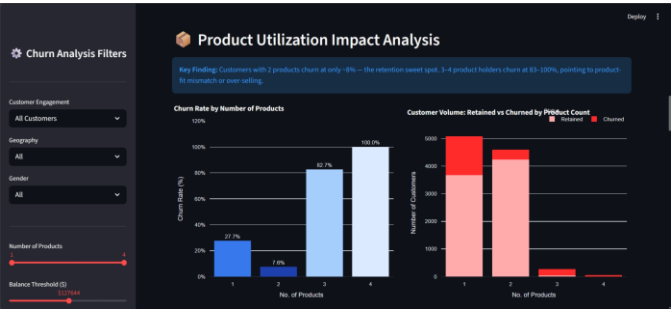


Fig. 2. Product Utilization Impact Analysis

This section examines the relationship between customer retention and the quantity of banking products they use. The results show that while higher product counts may indicate ineffective cross-selling or product misalignment, customers with moderate product adoption exhibit lower churn rates.

3)High-Value Disengaged Customer Detector:



Fig. 3. High-Value Disengaged Customer Detector

Customers with high financial value but low engagement are identified by this module. Despite having solid financial profiles, these clients constitute a high-risk market. Targeted retention strategies for premium clients are supported by this insight.

4)Retention Strength Scoring Panel:



Fig. 4. Retention Strength Scoring Panel

A relationship strength scoring model that incorporates engagement, product usage, and credit card ownership is part of the dashboard. In order to help prioritize retention efforts, this panel illustrates how churn varies across various relationship strength levels.

5)Interactive Controls and Filters:

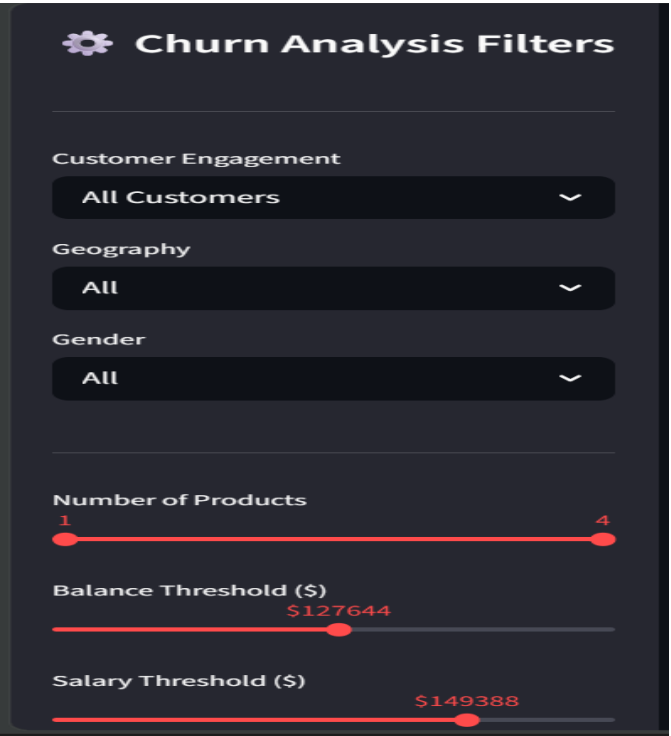


Fig. 5. Interactive Controls and Filters

The dashboard includes interactive elements like balance/salary thresholds, product count sliders, and engagement filters. The dashboard is very adaptable for business use thanks to these controls, which enable users to carry out drill-down analysis and dynamic segmentation.

In order to improve customer engagement and loyalty, stakeholders can identify at-risk customers, assess retention drivers, and make data-driven decisions thanks to the dashboard's real-time analysis capabilities.

IX. KEY INSIGHTS

The study gives a number of important insights into the factors that influence customer retention, with a focus on behavioral patterns rather than conventional financial metrics. The most important predictor of retention was found to be customer engagement; customers who were actively involved showed significantly lower churn risk than those who had little or no interaction.

Customers who use a moderate number of banking products—especially two products—show the highest levels of loyalty, according to the study. This implies that while excessive or misaligned product offerings might not always improve retention results, balanced product adoption strengthens the customer-bank relationship.

The identification of high-balance customers with low engagement as a high-risk segment is another important realization. These clients exhibit "silent churn" traits despite having solid financial profiles, suggesting that financial value by itself cannot ensure retention in the absence of regular interaction.

Additionally, customer segmentation based on churn risk and loyalty was successfully achieved by using a composite relationship strength score. While customers with moderate scores showed better retention stability, those with lower relationship strength were much more likely to churn.

All things considered, these results highlight the significance of using behavioral analytics to create more focused and successful retention strategies, allowing financial institutions to go beyond general methods and concentrate on significant client interactions.

X. STRATEGIC RECOMMENDATION

Several strategic recommendations can be put into practice to increase customer retention and fortify long-term relationships based on the analysis's findings. Banks should concentrate on creating targeted engagement programs for inactive or low-activity customers, viewing customer engagement as a key retention lever. Re-engaging these customers and lowering their churn risk can be achieved through proactive communication, tailored offers, and more digital touchpoints.

Banks should also implement more organized and customer-focused product bundling strategies. Customer relationships can be strengthened by encouraging customers to use complementary financial products, such as combining savings accounts with credit cards or digital banking services. However, in order to guarantee meaningful usage rather than excessive

cross-selling, product recommendations should be in line with customer needs.

High-value clients, especially those with large balances or income levels, should receive extra care. Reduced activity may be a sign of early churn, so it's important to keep an eye out for engagement signals from these clients. Retaining these financially valuable segments can be aided by the implementation of early warning systems and focused interventions.

Lastly, banks should use data-driven personalization to improve their relationship management procedures. Over time, better retention results can be achieved by greatly improving customer experience and loyalty through tailored communication, specialized financial solutions, and committed support for premium clients.

XI. CONCLUSION

This study shows that behavioral factors like engagement and product utilization, rather than just traditional financial indicators, are what drive customer retention in the banking industry. The results show that while behavioral analytics offers deeper and more useful insights into retention patterns, depending only on metrics like balance or income may not give a comprehensive picture of customer loyalty.

The project creates a workable and scalable framework for identifying important retention drivers and tracking customer behavior by combining exploratory data analysis with an interactive dashboard. By using this method, financial institutions can improve their capacity to identify at-risk clients and take prompt action by moving toward data-driven, engagement-focused retention strategies.

Additionally, using dynamic visualization tools makes insights more accessible to business stakeholders, facilitating more proactive and informed decision-making. Predictive machine learning models can be added to this analysis as a next step to predict customer attrition. By anticipating churn risk ahead of time and implementing focused interventions, these models would help banks increase customer retention and long-term value creation.

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